

Private Databases on the Cloud: Models, Issues and Research Perspectives

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Abstract—Privacy and security of big data is emerging as one among the most relevant research challenges of recent years, also stirred-up by a wide family of critical applications ranging from scientific computing to social network analysis and mining, from data stream management to smart cities, and so forth. Traditionally, the issue of making (even very-large) databases private and secure has a long history in the context of encrypted databases but, when specifically considered in the Cloud setting, it poses new requirements and challenges to deal with, with particular regard to the scalability of solutions. In line with this emerging research trend, this paper focuses the attention on state-of-the-art proposals in the area of private databases over Clouds, and proposes critical comments about pros and cons of actual research efforts along with future research directions to be considered in future years.

Keywords—Big Data, Privacy and Security of Big Data, Private Databases on the Cloud, Encrypted Cloud Databases

I. INTRODUCTION

Nowadays, Clouds are among the most challenging computational infrastructures of interest for actual research communities. Clouds not only pose architectural and infrastructural issues but also paradigm-oriented research challenges. Among these, *privacy and security of big data on the Cloud* play, without doubts, a dominant role (e.g., [13], [27]). This happens, for instance, in the so-called *hybrid Clouds* (e.g., [37]), which are emerging as one of the most popular Cloud environments for modern applications, such as *energy-efficient computing* (e.g., [5]), due to the fact that, in such Clouds, *outsourced databases* (e.g., [28]) are very often introduced and managed.

Indeed, it is easy to note that a plethora of Cloud-based applications, such as *scientific computing*, *social network analysis and mining*, *data stream management*, *smart cities* and so forth, expose to the requirement of accessing, managing and processing big data over Clouds. From this, it is natural to derive the need for models, techniques and algorithms for defining *privacy-preserving database management solutions over Clouds*, with *scalable extensions* towards more specific big data processing. It is a matter of fact, indeed, the strong correlation between Cloud architectures and big data management (e.g., [3]).

Private databases on the Cloud introduce a wide family of research issues. One is represented by the issue of *securely accessing a Cloud database*, which has attracted a relevant amount of attention from the research community (e.g., [29]).

Basically, here the problem consists in providing effective and efficient secure access schemes to Cloud databases, by also ensuring a well-recognized features like: *grant management*, *grant revocation*, *dynamic access*, and so forth. Another relevant issue is represented by the requirement of making Cloud databases *privacy-preserving*, meaning that they must ensure the privacy of data during common (data) management tasks like *indexing*, *query processing*, *information retrieval*, and so forth.

The accurate management of private databases over Clouds is essential also because Cloud databases are the *target* of main data-intensive processes like *big data analytics* (e.g., [24], [12]). Analytics tasks that compose wider big data analytics processes must *repetitively* access and process databases on the Clouds, thus posing several challenges to deal with and achieving the so-called *secure big data analytics* research niche (e.g., [14]). The involved issues get worse when big data analytics are performed in *fully-distributed environments*. As strictly related to Cloud architectures, an interesting line of research for supporting private databases is represented by *virtualization techniques* (e.g., [44]). Just like other resources, even databases can be virtualized over Clouds with the additional requirement of adding specific private and secure features to them (e.g., [34]).

A significant line of research that has recently emerged as “one solution to all issues” trend is represented by the so-called *encrypted databases* (e.g., [36]), which, however, is not new in the research community. Basically, this paradigm predicts to encrypt database tuples (according to state-of-the-art encryption algorithms – e.g., [23]) in order to enforce their privacy and security. This immediately demands for introducing models, techniques and algorithms for supporting common database management tasks, such as indexing, cleaning, querying and so forth, over *encrypted data*, yet having available ad-hoc routines to move from the encrypted domain to the decrypted domain, and viceversa.

An interesting application scenario for private databases in the Cloud is represented by the *Internet-of-Things* (IoT – e.g., [40]) environment. Here, *smart applications* make things interacting and inter-operating via ad-hoc wireless network protocols, and are foreseen to access and query device-embedded data, yet posing privacy and security challenges. The integration among IoT platforms, Clouds and encrypted databases is a very promising research line to be considered by future research efforts.

From this analysis, it clearly follows that a lot of research must still be done in the context of private databases on the Cloud, perhaps by extending models and solutions achieved in related research contexts like secure access models and encrypted databases. On the basis of this main motivation, this paper focuses the attention on state-of-the-art proposals in the area of private databases over Clouds, and proposes critical comments about pros and cons of actual research efforts along with future research directions to be considered in future years.

The remaining part of the paper is organized as follows. Section II provides a brief overview on hybrid Clouds that, as highlighted above, are one of the most relevant Cloud-based architecture for which private database management plays the major role. Section III provides an overview on some recent relevant proposals falling in the context of private databases over Clouds. Section IV contains our analysis and critical comments on next-generation techniques for supporting private databases over Clouds that, in our opinion, will represent a milestone for future research efforts. Finally, Section V provides the conclusions of our work.

II. HYBRID CLOUDS: MODELS, DEFINITIONS AND A REFERENCE ARCHITECTURE

Hybrid Clouds are a special case of *inter-Clouds* (e.g., [10]) where private and public Clouds are interconnected and integrated. The *International Data Corporation* (IDC) study predicted that, in 2015 [2], chief information managers will move to hybrid Cloud and, as part of this migration, existing deficiencies in service management will become evident, forcing investment in automation and consumption of externally managed services as alternatives to on-premises deployment of Cloud. Key trends that will emerge in the Cloud market include demand for integrated software development methods that stress communication, collaboration, integration, automation and measurement of cooperation between software developers and other IT professionals. There is also demand for higher adoption of virtualization solutions and increased emphasis on operational governance and security management. Among these challenges, privacy of databases “disseminated” in the Cloud is also comprised.

On March 24th, 2014, Cisco announced plans to build the world’s largest global Inter-Cloud together with a set of partners to create a network of Clouds called *Inter-Cloud Fabric* [1]. Cisco’s Inter-Cloud strategy will essentially facilitate the move toward hybrid Clouds and potentially beyond. Cisco is planning to spend 1BN on Inter-Cloud over the next couple of years, and has more than 3,700 people working on it. Their partners will be connecting more than 400 Data Centers (DC) to Inter-Cloud collectively in 50 countries. To enable IT and facilities side convergence and flexibility, Cisco will need to partner with suppliers that are developing advanced versions of *DC Infrastructure Management* (DCIM), which is also classified as *DC Services Optimization* (DCSO) in the Inter-cloud scenario.

As a reference architecture of hybrid-Cloud systems, in the following we provide a description of *EcoMultiCloud* [18], a *hierarchical framework for the efficient distribution of the workload on a multi-site scenario among geographically-distributed interconnected DCs*. It allows for an integrated

and homogeneous management of heterogeneous platforms but at the same time it preserves the autonomy of single sites. Through the self-organizing and adaptive nature of the approach, the Virtual Machines (VM) migrations are performed asynchronously, both location-wise and time-wise, and with a tunable rate managed by DC administrators.

The *EcoMultiCloud* hierarchical architecture is composed of two layers:

- **The lower layer** is used to allocate the workload within single DCs: each site adopts its own strategy to assign VMs internally, with local consolidation algorithms (possibly different from site to site). The lower layer collects information about the state of the local DC, and passes it to the upper layer.
- **The upper layer** is able to exchange information among geographically dispersed and interconnected sites and drive the distribution of VMs among the DCs.

Workload management in a geographical scenario is typically solved as an optimization problem, often in a centralized way. This approach has three main implications: (i) poor scalability, due to the large number of parameters and servers; (ii) poor ability to adapt to changing conditions, as massive migrations of VMs may be needed to match a new decision on of the workload distribution; (iii) limitation to the autonomy of the sites, which are often required to share the same strategies and algorithms. To tackle these challenging issues, the *EcoMultiCloud* solution is to design and develop a hierarchical framework for the efficient distribution of the workload on a multi-site platform. To the best of our knowledge, the *EcoMultiCloud* approach is the first to offer a solution for the multi-DC scenario that exploits the benefits of a hierarchical architecture, balances multiple business objectives and constraints, and integrates algorithms for the assignment/routing problem and algorithms that trigger inter-DC migrations to adapt the workload distribution to varying conditions. The *EcoMultiCloud* hierarchical approach does not cause performance degradation with respect to single layer algorithms, and it additionally offers notable advantages in terms of time to convergence (because the bigger problem is decomposed in several smaller ones), scalability, autonomy of sites, overall administration, information management.

The reference scenario is depicted in Figure 1, which shows the upper and lower layer for two interconnected data centers, as well as the main involved components. At each data center, a *Data Center Manager* (DCM) runs the algorithms of the upper layer, while the *Local Manager* (LM) performs the functionalities of the lower layer. Three basic algorithms must be designed and implemented at each DCM: (i) the *assignment algorithm* that determines the appropriate target DC for each new VM; (ii) the *migration algorithm* that determines from which source site and to which target site the workload should be migrated; (iii) the *redistribution algorithm* that periodically evaluates whether the current load balance is appropriate and, if necessary, decides whether an amount of workload should be migrated to/from another site. The assignment algorithm is used to route a new VM to the best target DC. However, the workload distribution may become inefficient when the conditions change, e.g., the overall load or the price of energy

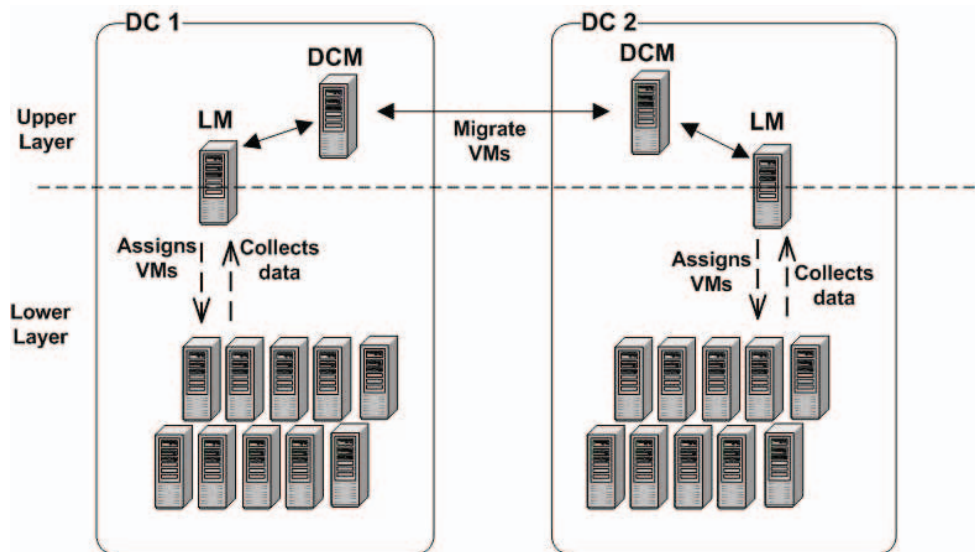


Fig. 1. EcoMultiCloud Hierarchical Architecture

may vary in one or more data centers. In such cases inter-DC VM migrations are performed to redistribute the workload considering the new conditions.

Due to the rapid emerging of the Hybrid Cloud paradigm, a hierarchical architecture like the one devised with the *EcoMultiCloud* solution, will need to be adapted to scenarios in which some data centers are private and some others are public and offered by Cloud infrastructures. For example, one of the two DCs in Figure 1 can be an on-premises data center and the other one a Cloud data center. It is clear that the management of private data is crucial in this context, as VMs, and the related data, can be migrated from the private area to the public data center, and viceversa, in accordance with the defined technical and business goals.

III. STATE-OF-THE-ART TECHNIQUES FOR SUPPORTING PRIVATE DATABASES OVER CLOUDS

The research community has devoted quite a relevant amount of attention to the issue of supporting private databases over Clouds. In the following, we provide an overview on some recent relevant proposals.

[19] focuses the attention on the problem of *supporting private Nearest-Neighbor (NN) queries over databases* in the context of distributed mobile environments, which well-marry with the case of (mobile) Clouds. In particular, authors are interested in supporting the privacy of users via hiding their exact user coordinates to un-trusted entities that are delivering mobile services associated to *Points of Interest (POI)*. To this end, a two-step approach for *supporting private location-based query processing* is defined. At the first step, the locations of users are hidden inside so-called *Cloaking Regions (CR)*; at the second step, location data (which are embedded in mobile databases) are encrypted by using a suitable *Private Information Retrieval (PIR)* protocol (e.g., [41]). In addition to this, ad-hoc algorithms for supporting approximate and exact NN queried are provided and experimentally assessed.

[42] moves the attention to a topic that is related and,

also, extends the previous one. Authors consider the interesting applicative setting where the access and the processing of private queries over encrypted databases must be performed not just for a *single user* (like the majority of proposals do) but, rather, for *multiple users simultaneously* (hence, the encryption scheme cannot be designed for the support of a specific class of queries). The approach is relevant in practice because, as authors correctly state, most databases in practice do not just serve one user but, instead, they support search and write operations by multiple users. In order to provide the described features over private databases, authors propose a set of *security notions for multi-user searchable encryption* and, in addition to this, the proof that such an encryption scheme is provably secure under the newly-introduced security notions.

[26] shows the versatility and the flexibility that must characterize private databases over Clouds by focusing on the issue of *supporting private analysis of graph databases*. It should be noted that this challenge is very interesting for modern big data analytics systems. The idea underlying this research effort consists in releasing *useful statistics* about graph data stored in the target database while providing rigorous privacy guarantees. With this goal in mind, authors provide definition and experimental analysis of two algorithms that output approximate answers to *sub-graph counting queries* (e.g., [32]). In particular, these algorithms retrieve the number of edge-induced isomorphic copies of an input query graph, by introducing a new class of statistics capable of nicely describing the nature of target graph database, called *k-star queries* and *k-triangle queries*, respectively. In the experimental evaluation, authors prove that their proposed algorithms outperform state-of-the-art proposals, by testing them on both real-life and synthetic data sets.

[39] addresses a problem that is very relevant at now, i.e. *making private queries against a public database* (e.g., Web-available government databases, public statistical databases – e.g., census data, and so forth). It should be noted that this challenge is strictly related to the emerging *linked open data* context (e.g., [20]) and it can be perfectly integrated in Cloud

infrastructures (e.g., [6]). Authors propose two protocols for private processing of database queries whose final goal is that of hiding contents of user queries and user data to online public service providers. These protocols, called BHE and HHE, respectively, make use of the well-known *Pailliers homomorphic encryption scheme* (e.g., [31]), and they are capable of supporting a rich class of user queries (including range queries, join queries, and so forth) against public databases. Specific characteristics of BHE and HHE are different. BHE is a fundamental private query processing protocol for public data sets that still incurs in high communication costs among database clients and public databases. HHE is an extension of BHE and predicates the application of *ciphertext computation* (e.g., [30]) over frequently-accessed partitions of data. Privacy of users according to HHE is theoretically and experimentally proved in the paper.

[43] proposes a variant of the well-known *SARG04 protocol* (e.g., [9]), which supports private query processing via generating an oblivious key that then allows user to retrieve one item from the target database without publicly revealing which is the item of his/her interest. Indeed, it has been proved that private query processing protocols that found on SARG04 as basic protocol are vulnerable to *faked data attacks*. Hence, in order to deal with this drawback, authors propose the new SARG04 variant that allows the user's privacy to be significantly improved. In more detail, according to the new protocol, an *honesty test* (e.g., [21]) is used to detect a cheating database that has transmitted faked data during previous sessions. Authors also experimentally prove that the proposed protocol is efficient in terms of communication costs.

[17] considers again distributed settings as reference application scenario and studies the problem of *supporting efficient and private approximations of distributed databases calculations*. Authors depict a target environment where data are collected in *different and non-cooperative databases*. For such specific settings, the issue of supporting privacy-preserving distributed calculations over these kind of databases has been largely investigated. While some existent approaches exist, computational costs are still a terrible bottleneck to deal with. Inspired by this main evidence, authors provide an innovative solution that *trades-off between performance and accuracy* of distributed calculations via *data sampling* (e.g., [22]). They focus on the specific case of supporting sampling for separate, non-collaborating, vertically-partitioned data sets, yet supporting privacy-preservation requirements. In order to provide a proof-of-concept, authors apply the proposed method to approximation of intersection set both without and with privacy-preserving mechanism, and derive an interesting theoretical analysis on the error bound that, along with the experimental assessment and analysis of the proposed method, completes the contributions of the paper.

[25] investigates a problem that is relevant in practice, i.e. supporting *Searchable Symmetric Encryption* (SSE) (e.g., [11]) over private large-scale databases. By simplifying, this means that users can still apply search operations over encrypted data, without lack of effectiveness and efficiency for the majority of supported search types. As authors correctly recognize, the main issue for SSE schemes is represented by the fact that a limited kinds of search procedures are available to users accessing an SSE-encrypted database. In order to fulfill this

gap, authors propose an approach where *the SSE scheme is built on top of a B-tree* supporting a wide variety of search features (as, for instance, range queries, sub-string queries, and so forth), and the overall method is made available to users in the vest of two combined Cloud services. The so-constructed index is managed and maintained by smoothly trading-off privacy and efficiency. An extensive experimental work and related assessment confirms the performance of the private database solution.

[38] considers the application scenarios where users/applications demand for the execution of *private SQL aggregates on a secure server*, being such processing scalable on large-scale distributed environments. It should be noted that this paradigm meets the principled concepts of Cloud data processing. Authors recognize the dichotomy between actual applications collecting enormous quantities of personal information and the possible privacy breaches of the a central server where such information are finally stored. In order to face-off this problem, authors propose to pushing the security to *secure hardware devices* controlling the data at the place of their acquisition, and, on top of such physical components, executing SQL aggregates without revealing any sensitive information to the central server. They also study how to secure the execution of such queries in the presence of *honest-but-curious* ([35]) and malicious attackers. A complete reference decentralized architecture and a comprehensive experimental campaign, which truly proofs the scalability of the proposed framework, finalize the paper's contributions.

Finally, [8] presents *eSkyline*, a prototype system and query interface that enables the processing of *skyline queries over encrypted data*, even *without* preserving the order on each attribute as *order-preserving encryption* (e.g., [4]) would do. The proposed system includes an encryption scheme that facilitates the evaluation of *domination relationships* [7], hence allowing state-of-the-art skyline processing algorithms to be used. In order to prove the effectiveness and the reliability of *eSkyline*, authors also provide the details of the underlying encryption scheme, plus a suitable GUI that allows a user to interact with a server, and showcases the efficiency of computing skyline queries and decrypting the results.

IV. PRIVATE DATABASES OVER CLOUDS: FUTURE RESEARCH PERSPECTIVES

Several research issues will play a prominent role in next-generation techniques for supporting private databases over Clouds. In the following, we report on some of most noticeable of them.

Indexing Data Structures for Private Cloud Databases When dealing with private Cloud databases, the issue of defining and building *effective and efficient indexing data structures* enabled with data management features (e.g., searching a certain item) is very relevant. Indeed, the encryption task must not prevent the support of such operations. This need, combined with the obvious scalability requirement, is a clear research challenge of future efforts.

Advanced Query Operators over Private Cloud Databases Querying private Cloud databases not only implies the support of basic query operators (e.g, SELECT, UPDATE, and so forth), but also *advanced query operators* such as: aggregate

operators, rank-based operators, IR-style operators, OLAP-style operators, and so forth. The issue of supporting this rather-wide family of query operators on top of encrypted databases is significant as well, especially when these operators are used as “basic core layer” for more complex big data analytics tools.

Dealing with Mixed-Encrypted Cloud Databases Database encryption usually works at a certain *granularity* that, for instance, can be attribute-level or tuple-level. Both have advantages and disadvantages. For instance, attribute-level encryption may be useful during JOIN query processing, while tuple-level encryption ensures a more-compact and better-for-access-procedures solution. Starting from these considerations, an important line of research for private Cloud databases will be represented by the issue of introducing and devising *mixed-encryption schemes*, i.e. encryption schemes that incorporate different granularities, which may turn to be extremely useful in complex analytical settings.

Preference-Aware Query Processing over Encrypted Cloud Databases Preference-aware query processing (e.g., [33]) is a fortunate research niche within the database community. It foresees paradigms according to which users can embed so-called “preferences” in their query interaction, and the query processor filters-out the results by pruning those tuples that do not satisfy those preferences. Skyline queries are a kind of preference-aware query. While skyline query processing over encrypted data has been studied recently (e.g., [8]), to the best of our knowledge there are not relevant research efforts in the context of preference-aware query processing over encrypted data. We recognize the latter as one of the most prominent research challenge for private Cloud databases.

Distributed Query Processing over Private Cloud Databases With the goal of supporting big data analytics tasks, query processing over private Cloud databases usually performs in a *distributed manner*, i.e. private databases stored in *different nodes* of the reference Cloud architecture must be accessed and queried. This calls for solutions capable of devising *distributed query optimization plans* in the presence of encrypted data. A possible alternative is represented by the idea of trading-off accuracy of the query answers with the privacy of target Cloud databases. *Adaptive query optimization plans* are also interesting to this purpose.

Scalability Issues One of the requirements of private Cloud databases is, without doubts, *scalability*. Indeed, Cloud databases are used to run in complex environments where performance plays a leading role. To this end, not only the specific query algorithms, which are likely to run online, but also the encryption algorithms, which are likely to run offline, must marry *elastic metaphors* and scale-up on massive big data. This is still an open issue to be widely investigated, as it heavily impacts on the reliability of real-life big data applications and systems.

Integration with MapReduce Engines MapReduce [16] is a novel computational paradigm according to which complex procedures are *decomposed* in simpler sub-procedures (the *map* phase) and later the final computation result is *composed* by combining the local results of such sub-procedures (the *reduce* phase). This paradigm is likely to be integrated with private Cloud databases in a smoothly way, even with the goal of

computing the encrypted versions of massive Cloud databases in a parallel/distributed-manner. The integration among private Cloud databases and MapReduce engines is really likely to become one of the most interesting and exciting research avenues in the field of big data processing.

Integration with IoT Architectures When integrated with IoT architectures, private Cloud databases expose a *higher risk of privacy breaches*. This is due to several reasons. First, devices are characterized by very heterogeneous drivers and communication protocols. Second, several devices do not have an “explicit” Internet connection protocol. Third, for some devices (e.g., those used in *domotics applications*) the mobility is really high, hence this complicates the security- and privacy-preserving process. All these concepts are likely to be further studied in next research efforts.

Encryption Mechanisms over Uncertain Cloud Databases Nowadays, dealing with privacy-preserving issues over *uncertain data sets* (e.g., [15]) is a gold research topic for the database community. Uncertainty can be found in a plethora of real-life application scenarios, hence from this evidence several research efforts have been devoted recently. With the same spirit, this problem will soon touch the field of encryption mechanisms for Cloud databases. Hence, devising models, algorithms and techniques for dealing with *encryption mechanisms over uncertain Cloud databases* is regarded as a rich line of research for the future years.

V. CONCLUSIONS

Recently, a great deal of attention has been devoted to private databases over Clouds, as stirred-up by both (i) interesting theoretical problems and challenges and (ii) novel applications where secure and privacy-preserving access and management of embedded databases pose strict requirements, like, for instance, the case of emerging IoT architectures. In response to this “call for arms”, a lot of research efforts has been devoted to design and implement a wide spectrum of solutions and achievements. Inspired by this trend, this paper has focused the attention on state-of-the-art proposals in the area of private databases over Clouds. A critical evaluation of current research efforts along with perspectives on future research directions have been proposed as well.

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